



The Antarctic Ice Sheet

The world's largest glacier, having a volume of some 7.2 million cubic miles (30 million cubic km) and holding over 60 percent of Earth's freshwater

The Antarctic Ice Sheet is by far the most extensive continuous mass of ice on Earth, covering an area of 5.4 million square miles (14 million square km). It consists of two main, neighboring, parts. The larger of these, the East Antarctic Ice Sheet, is over $2\frac{3}{4}$ miles (4.5 km) thick in places and lies on top of a large landmass. It has an area of about 4.6 million square miles (12 million square km). In contrast, the smaller West Antarctic Ice Sheet has an area of about 0.8 million square miles (2 million square km) and a maximum thickness of $2\frac{1}{4}$ miles (3.5 km), and rests on bedrock that is mostly below sea level. Both parts are slightly dome-shaped, with ice flowing from their

highest parts toward the coasts. The rate of ice movement varies from less than 3 feet (1 m) per year in some of the higher regions to a thousand feet (several hundred meters) per year in outlet glaciers and ice streams that drain ice down to the coasts. In some places, the outlet glaciers merge to form vast platforms of floating ice called ice shelves, or extend over the sea as smaller ice tongues.

Transantarctic Mountains
Along a section of their length, the mountains form a boundary between the two main parts of the ice sheet.

► THE ANTARCTIC ICE SHEET

The ice sheet, shown here with a cross-section through the Ross Ice Shelf, covers about 99 percent of the Antarctic landmass. The only ice-free areas are the peaks of the highest mountains and a few coastal areas of limited extent.

► SPEEDY GLACIER

The massive Byrd Glacier, which drains part of the East Antarctic Ice Sheet into the Ross Ice Shelf, is about 84 miles (136 km) long and 15 miles (24 km) wide. Its flow rate is high for a glacier—about $\frac{1}{2}$ mile (750 m) per year.

► ICE FLOES

Sea ice occurs all around Antarctica, although its extent varies seasonally. Unlike ice shelves and icebergs, which originate on land, sea ice is frozen seawater and contains some salt. Flat, freely moving sheets of sea ice are called floes.

Southern Ocean surrounds Antarctica on all sides

the Antarctic Peninsula is mountainous and about 80 percent is ice covered

West Antarctica

Lake Ellsworth

This is one of many subglacial lakes. It is buried under about $2\frac{1}{8}$ miles (3.4 km) of ice.

the Ronne Ice Shelf is the world's second largest ice shelf

